Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C013 **CYC3** 3 genes / 2 species 2 direct pairs	**Y1 CYC3** GenAge 1414 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043 **Y2 CYT2** GenAge 1047 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 19030232 20657825	**W1 cchl-1** GenAge 94 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12447374, 16103914	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m); W1–Y2 (1:m)				
C123 **CYS4** 2 genes / 2 species 1 direct pair	**Y1 CYS4** GenAge 1569 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20883526	No gene in this candidate network	**F1 Cbs** GenAge 1791 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21930912	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				
C018 **CYT1** 2 genes / 2 species 1 direct pair	**Y1 CYT1** GenAge 1701 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043	**W1 cyc-1** GenAge 145 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16103914, 20688098	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C132 **dgk-5** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 dgk-5** GenAge 1931 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24889782	**F1 rdgA** GenAge 1929 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24889782	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.